tandemo

True tandem-repeat length distributions per locus and haplotype
from PacBio HiFi reads: a learned read likelihood and a soft-phased mixture

Method document (software version 0.1.0)

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Abstract

Observed read-length variation at a tandem repeat mixes sequencing noise with real variation between molecules. **tandemo** separates the two in two stages. *Network 1* learns, on a sample without repeat instability, how a molecule of known length L turns into an observed repeat length Y in a HiFi read, given the read's quality and the repeat's context: $p(Y \mid L, x)$. TRGT supplies the germline lengths, and HiPhase the phase evidence. *Stage 2* then estimates, for every locus and haplotype of a new subject, the distribution of *true* lengths $f_h(L)$: a point mass at the germline length plus a variable component. It integrates over each read's unknown length and assigns reads to haplotypes softly, by het-SNV evidence along the whole read. The output is one row per subject, locus and haplotype, ready for group comparison.

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1 Overview

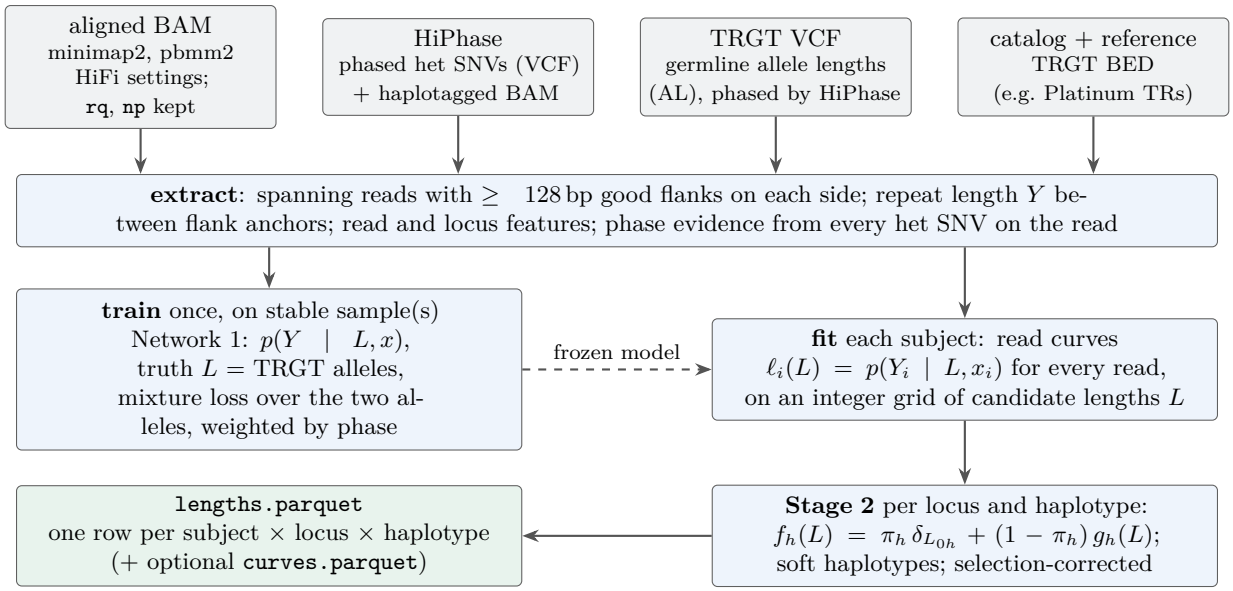


Figure 1: The pipeline. Network 1 is trained once, on one or more stable samples, and then frozen. Every subject is extracted and fitted on its own.

The central distinction (from the framework this implements) is between *measurement* and *biology*:

- **Measurement**, Network 1: if a molecule were L bases long, how likely is each observed length Y in this read? This depends on the read and the repeat, never on the subject's biology.
- **Biology**, Stage 2: how are true lengths distributed among the molecules of one haplotype? Each read's own length is unknown and is summed out, never assigned.

2 From aligned reads to observations

2.1 Which reads, and how Y is measured

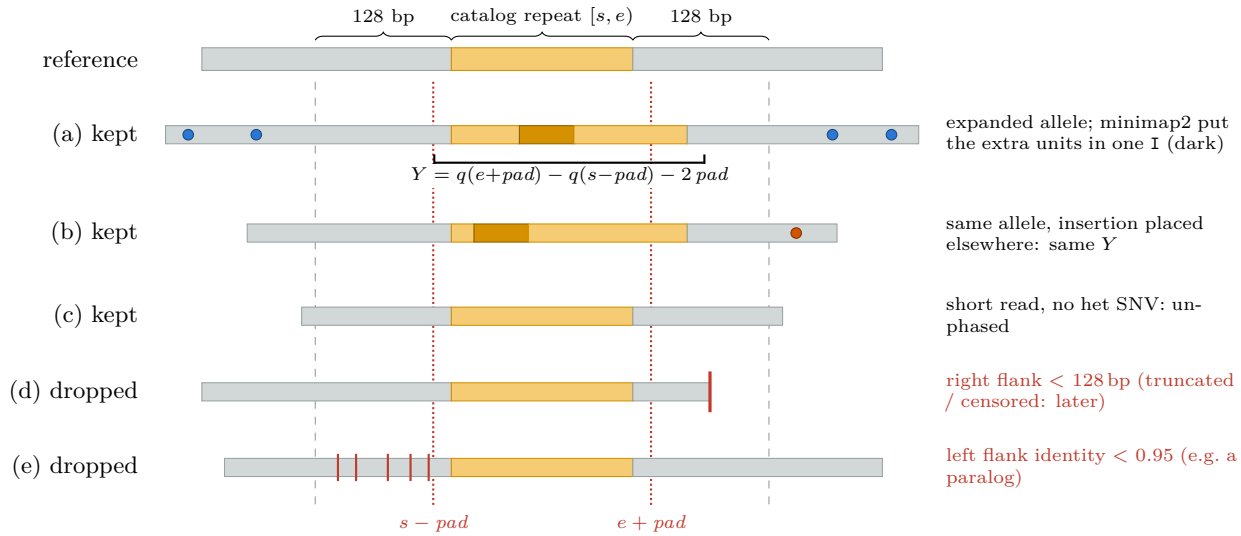


Figure 2: Reads at one locus. A read is kept when it covers 128 reference bases on *each* side of the repeat, with ≥ 95% alignment identity there and no clipping. Y counts every read base between two anchors in the unique flanks (default $pad = 10$), including bases minimap2 called insertions. Reads (a) and (b) come from the same expanded allele: minimap2 placed the insertion at different points inside the repeat, but Y is identical. Coloured dots are phased het SNVs anywhere on the read (blue haplotype 1, orange haplotype 2).

Why measure between anchors? Inside a repeat, minimap2's choice of where to place an indel is arbitrary. Two reads of one allele can carry the extra units at different positions, and gap penalties sometimes trade an

indel for mismatches. The anchors sit outside the catalog interval (plus *pad*) in unique flank sequence. Every indel the aligner placed inside the repeat is therefore counted wherever it went, and repeat edges that blur into partial units are measured consistently in every read.

Read filters. Primary alignments only (no secondary, supplementary, duplicate or QC-fail records), MAPQ ≥ 20, HiFi read quality `rq` ≥ 0.99. Truncated reads, which end inside the repeat, are not used yet (Section 7).

2.2 Phase evidence along the whole read

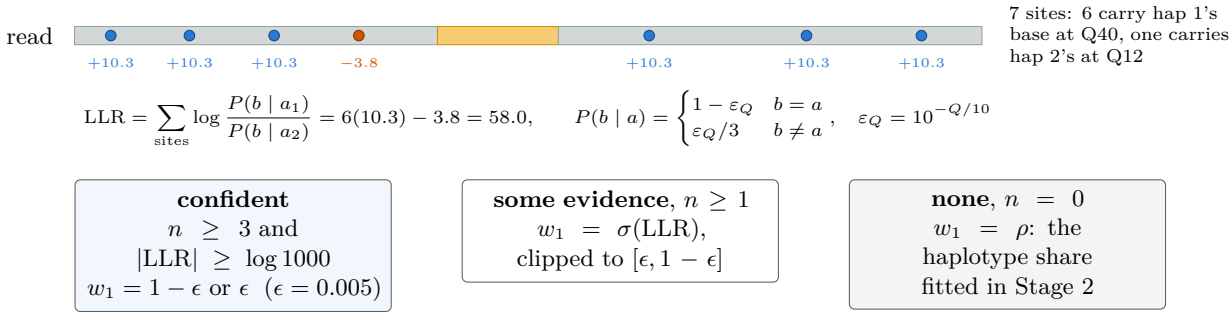


Figure 3: Phase weight of a read. Every phased het SNV the read covers counts, not only those in the flanks. Only sites in the locus's phase block (the PS of the phased TRGT record, or of the nearest het sites) count. Het indels count too, when the read's bases from 10 bp before to 10 bp after the indel match one haplotype's sequence exactly: they add $\pm \log \frac{1-e}{e}$ with $e = 0.01$, or $e = 0.05$ in a homopolymer run ≥ 4, where HiFi's residual errors concentrate. Indels within 5 bp of any catalog repeat are not used, since their alleles are repeat lengths. w_1 is a *prior* from germline evidence alone. Length never enters it, so length cannot be used circularly to hide instability. The cap ϵ keeps one misphased read from forcing an outlier onto the wrong haplotype.

2.3 Germline alleles from TRGT

TRGT reports each allele's length over the catalog interval $[s, e)$ (`FORMAT/AL`). When HiPhase has phased the TRGT VCF, allele k of the genotype belongs to haplotype $k{+}1$ of that block. Otherwise, for training, the orientation is taken from confidently phased reads: the orientation that puts each haplotype's median Y nearer its allele wins. Haploid calls (one AL value) give one haplotype.

Putting TRGT alleles on the scale of Y: the pads. Y spans more than the catalog interval: from $s - pad$ to $e + pad$, minus the 2 *pad* anchor bases. The two pads are fixed 10 bp windows immediately outside the repeat, $[s-10, s)$ and $[e, e+10)$, with no further slop. A germline indel in a pad changes every read's Y but not TRGT's allele, so without a correction network 1 would learn it as a systematic read error. Extraction therefore records, from the phased small-variant VCF, the net length change of the germline indels in the two pads on each VCF haplotype (d_1, d_2), and training uses

$$L_h = \mathrm{AL}_h + d_h,$$

oriented to haplotypes with the same phase evidence as the alleles. Indels further from the repeat lie outside Y and do not matter. A locus is left out of training when its pads cannot be resolved:

event	treatment
indel inside a pad (homozygous, or het in the locus's phase block)	added to L_h
insertion exactly at s or e (TRGT may or may not count it)	locus excluded (ambiguous)
deletion spanning s or e	locus excluded (ambiguous)
deletion crossing the outer pad edge $s-10$ or $e+10$	locus excluded (ambiguous)
het pad indel unphased, or in another phase block	locus excluded (unphased)

On HG002 chr4 (T2T, Platinum TRs), 1,446 of 503,547 genotyped loci (0.3%) carry a pad indel; 8,859 are excluded as ambiguous, mostly homopolymers and dinucleotides, and 91 as unphased.

3 Network 1: the read-generation likelihood

3.1 What it outputs

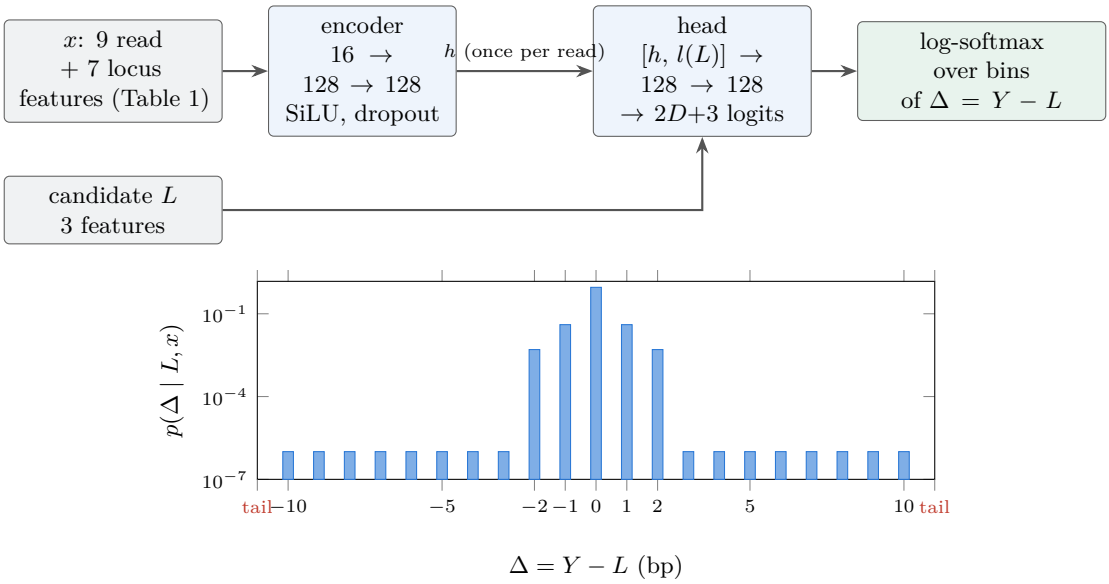


Figure 4: Top: architecture. The encoder runs once per read, and the small head once per (read, candidate L). That makes curves over hundreds of candidate lengths cheap. Bottom: the output for one read and one L is a distribution over the error bins: one bin per bp for $|\Delta| \le D$ (default 50), plus a left and a right tail bin whose mass is spread uniformly to $|\Delta| = T$ (5000). It sums to 1 over Y for each L, so it is a proper likelihood. It is *not* normalised over L, and must not be.

From outputs to a curve. For read i, the network is evaluated at every candidate L on the locus grid, reading off the probability of the read's *observed* Y_i: $\ell_i(L) = p(Y_i \mid L, x_i)$. This curve is all Stage 2 sees of the read (Figure 6).

3.2 Features

No feature encodes the read's own repeat length (no base counts or quality sums over the repeat), so the network cannot read Y off its input. The table is generated from `tandemo/features.py`.

group	feature	source column	transform	meaning
read	`rq_phred`	`rq`	Phred	HiFi read quality as Phred, -10 log10(1 - rq), capped at 60
read	`np_log`	`np`	log(1+x)	CCS passes, log(1 + np)
read	`read_len_log`	`read_len`	log10	read length, log10 bp
read	`mapq`	`mapq`	x/60	mapping quality / 60
read	`has_sa`	`n_sa`	$x > 0$	read has supplementary alignments
read	`bq_rep`	`bq_rep`	x/40	mean base quality over the read's repeat bases / 40
read	`bq_flank`	`bq_flank`	x/40	mean base quality over the two 128 bp flanks / 40
read	`dist_end_log`	`dist_end`	log(1+x)	bases from the repeat to the nearer read end, log
read	`flank_err`	`flank_id_min`	100(1-x)	error rate of the worse flank, 1 - identity, x100
locus	`period_log`	`period`	log2	motif period, log2 bp
locus	`ref_len_log`	`ref_len`	log(1+x)	reference repeat length, log bp
locus	`gc`	`gc`	x	GC fraction of the reference repeat
locus	`hp_frac`	`hp_frac`	x	fraction of repeat bases in homopolymer runs ≥ 4
locus	`max_hp_log`	`max_hp`	log(1+x)	longest homopolymer in the repeat, log
locus	`cx3`	`cx3`	x	3-mer complexity of the repeat (distinct / possible)
locus	`flank_cx`	`flank_cx`	x	3-mer complexity of the lower-complexity flank
candidate L	`L_log`	–	–	candidate true length, log(1 + L), standardised
candidate L	`L_vs_ref`	–	–	log(1 + L) - log(1 + reference length): distance of the allele from the reference
candidate L	`L_units_log`	–	–	log(1 + L / period): candidate length in motif units

Table 1: Inputs of Network 1. Missing values (e.g. no `np` tag) are filled with the training median before standardisation. The medians, means and standard deviations are stored in the model file.

3.3 Training on a stable sample

TRGT gives two allele lengths per locus. Which allele a training read came from is uncertain at heterozygous loci, so reads are never hard-assigned:

$$J = -\sum_i \log\left[\, w_{1i}\, p(Y_i \mid L_1, x_i) + (1 - w_{1i})\, p(Y_i \mid L_2, x_i) \,\right],$$

where w_{1i} is the read's phase weight (0.5 without evidence). Homozygous loci reduce to one term.

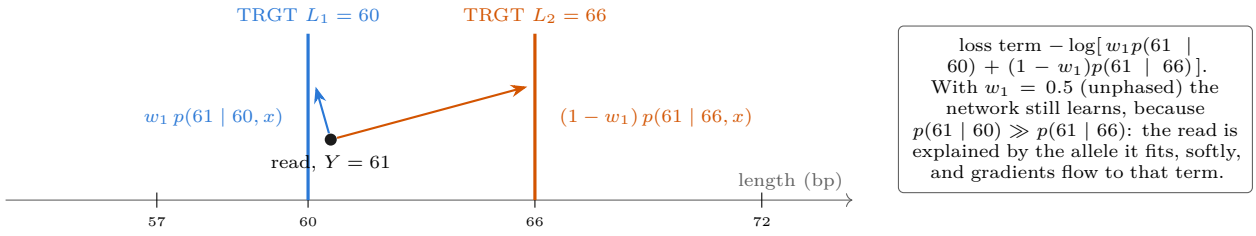


Figure 5: One training read at a heterozygous locus. The two TRGT alleles are both candidates. The phase weight and the network's own fit share the read between them, so no hard assignment is ever made.

setting	default	why
truth	TRGT `AL`	germline lengths of a sample without instability
loss	mixture NLL above	no hard allele assignment; uses heterozygous loci
validation	chr20–22 held out	reads of one locus never on both sides; else 10% of loci by hash
optimiser	AdamW, lr 10^{-3}, wd 10^{-5}, batch 4096	halve lr on plateau; early stop after 4 epochs
bins	$D = 50$, $T = 5000$	HiFi errors are mostly ± 1–2 bp; tails catch rare large errors
baseline	`--empirical`	histogram per period × length class, same loss (EM); no read covariates

Table 2: Network 1 training.

Cautions. TRGT's call is a consensus of the same reads, which pulls labels slightly toward them at low depth; a second run of the sample for labels avoids this. Calibrate on samples whose chemistry and allele-length range cover the subjects, since the network does not extrapolate far beyond its training lengths.

4 Stage 2: the distribution of true lengths

4.1 Combining read curves

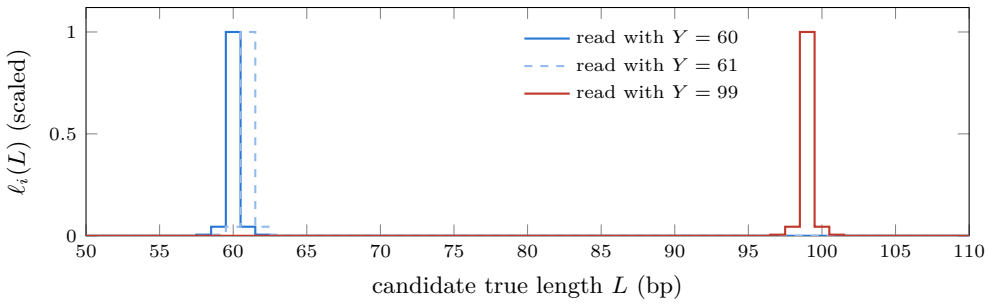


Figure 6: Three reads' likelihood curves from the noise model, scaled to peak 1. Each says which true lengths could have produced the read. Multiplying the curves assumes all reads share one length; that is model M0. Stage 2 instead averages each curve over a distribution before multiplying across reads.

For haplotype h, the true-length distribution f_h and the phase prior a_{ih} give each read the marginal likelihood

$$p_i = \sum_h a_{ih} \frac{\sum_L \ell_i(L)\, f_h(L)\, s(L)}{\sum_L f_h(L)\, s(L)}, \qquad a_{ih} = \begin{cases} (w_{1i},\ 1 - w_{1i}) & \text{het evidence} \\ (\rho,\ 1 - \rho) & \text{none} \end{cases}$$

The fit maximises $\sum_i \log p_i$ (computed with log-sum-exp) over the *shape* of each f_h and over ρ. No read is ever given a single length. Which haplotype a read belongs to comes out as its posterior

$$r_{ih} = a_{ih} \sum_L \ell_i(L) f_h(L) s(L) / Z_h \ / \ p_i,$$

which combines the het prior with the length fit. An unphased read whose Y sits at haplotype 2's L_0 lands there without any het SNV.

4.2 The model of f_h

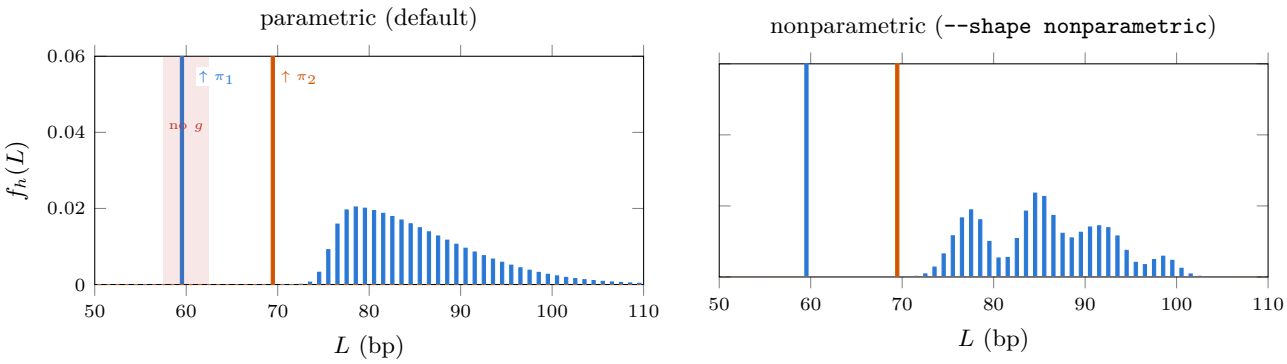


Figure 7: Fitted f_h on a simulated locus: haplotype 1 (blue) with 30% of molecules expanded around 84 bp, haplotype 2 (orange) stable at 69 bp; 45 reads per haplotype, 30% without phase evidence. Parametric fit: $L_{01} = 60$, $1 - \pi_1 = 0.31$ (truth 0.30), $1 - \pi_2 = 0.00$, log-likelihood gain over M0 = 135.3. Nonparametric: $1 - \pi_1 = 0.31$, $1 - \pi_2 = 0.00$. The point masses (arrows) run off the top: the axis is cut at 0.06 to show g. The shaded band is the g exclusion zone around L_0.

$$f_h(L) = \pi_h\, \delta(L - L_{0h}) + (1 - \pi_h)\, g_h(L), \qquad g_h(L) = 0 \ \text{ for } |L - L_{0h}| < \Delta_g.$$

- **Parametric** (default): g_h is a skew-normal(ξ, ω, α) evaluated on the integer grid and renormalised, with $\omega \geq \omega_{\min} = \max(1, \text{period}/2)$. That is seven numbers per locus and haplotype, stable at modest depth, and easy to compare across subjects.

- **Nonparametric**: g_h is a free distribution on the grid, fitted by EM with smoothing (EMS). Each iteration computes expected counts over L, undoes the selection $s(L)$, smooths with a Gaussian kernel (sd $\max(1, \text{period}/2)$), and restricts to the support. It finds bimodal or long-tailed shapes, at the price of noisier estimates.
- **Exclusion zone** $\Delta_g = \max(2, \text{period})$ bp. Without it, a narrow g sits on L_0 and absorbs ordinary ± 1 bp noise, making π unidentifiable (Section 6).

4.3 Fitting

1. **M0** ($\pi_h = 1$): coordinate ascent over (L_{01}, L_{02}) with ρ by EM. It starts from the phase-weighted modes, the TRGT alleles and the two largest pooled modes, each in both orders. The best is kept, so haplotype labels follow the phase evidence.
2. **M1 parametric**: L-BFGS-B over $(\operatorname{logit} \pi_h,\ \xi_h,\ \log(\omega_h - \omega_{\min}),\ \alpha_h)$ and $\operatorname{logit} \rho$. It starts from shapes on both sides of L_0, a broad shape, and the most likely length of each read M0 explains worst. Without those last starts, a single far-out read is never reached. Then a local rescan of L_0 (± 2 periods) and a polish.
3. **M1 nonparametric**: EMS from half a local bump and half a uniform floor over the support. EM cannot grow mass where it starts at zero.
4. If M1 ends below M0 (it contains M0), M0 is reported.

4.4 Spanning selection

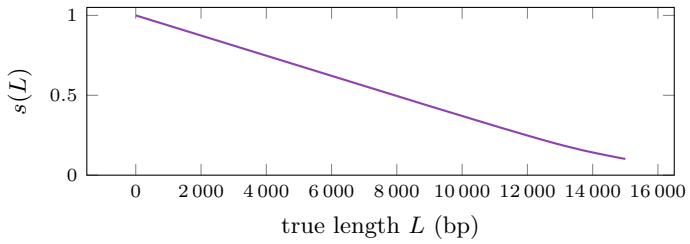


Figure 8: A molecule is only observed when a read covers it plus 128 bp on each side, so long molecules are undersampled. $s(L) \propto \sum_r \max(0, R_r - L - 2 \cdot 128)$ over the lengths R_r of all usable reads overlapping the locus (here around 16 kb). Stage 2 models the *kept* reads as $f_h(L)s(L)$, so f_h is the distribution of molecules. `--no-selection` turns this off.

5 Haplotypes and instability

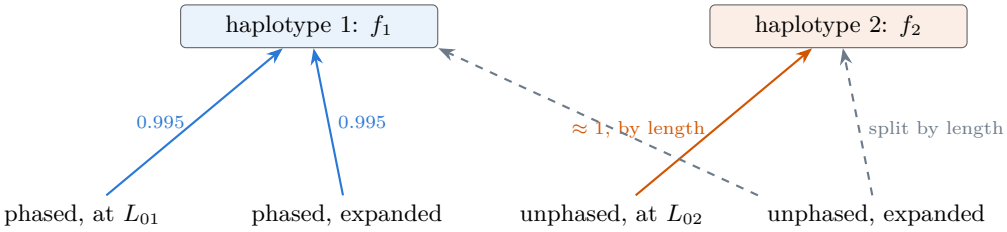


Figure 9: A somatic expansion in a haplotype-1 molecule keeps haplotype 1's het SNVs, so phased reads carry it to haplotype 1 only. Leakage between haplotypes happens only through reads without phase evidence, whose posterior is decided by length. Leakage moves variable mass between haplotypes but never creates it.

column	meaning
`L0`, `pi`, `var_weight`	germline length; point-mass weight; $1 - \pi$
`g_xi`, `g_omega`, `g_alpha`	skew-normal parameters (parametric; NaN for nonparametric)
`g_mean`, `g_sd`	moments of g
`p_exp`, `p_con`	$P(L \geq L_0 + \Delta)$, $P(L \leq L_0 - \Delta)$ under f_h, $\Delta = \max(3, \text{period})$, fixed in advance
`mean_abs_dev`	$E\|L - L_0\|$ under f_h
`n_eff`, `n_phased_eff`	reads received ($\sum_i r_{ih}$), of which with het evidence
`n_var_eff`, `n_support`	expected reads from g; reads whose posterior favours g (> 0.5)
`unphased_var_frac`	share of the variable component carried by unphased reads (attribution confidence)
`pooled_var_weight`, `pooled_p_exp`, `pooled_p_con`	locus level, $\sum_h \text{share}_h \times \cdot$: unaffected by leakage
`share`, `rho`	haplotype share; fitted ρ
`ll_m0`, `ll_m1`, `d_ll`	log-likelihoods and gain (evidence; no p-value)
`hap`, `hap_rank`	HiPhase label (arbitrary per block); `short`/`long`/`tie`/`haploid` by L_0
`converged`, `at_boundary`, `status`	fit diagnostics

Table 3: `lengths.parquet`: one row per subject × locus × haplotype. The full f_h can be rebuilt from the parameters (parametric). `--write-curves` also stores every read's curve near its peak (within 20 log units) with its phase weight, so Stage 2, or a group-level model, can be refitted without Network 1.

Group comparison. HiPhase labels are arbitrary per phase block, so haplotype 1 in one subject is unrelated to haplotype 1 in another. Compare by `hap_rank` (short/long allele), or use the pooled columns. Read all subjects as one Parquet dataset (DuckDB, polars, pandas) and join a sample table.

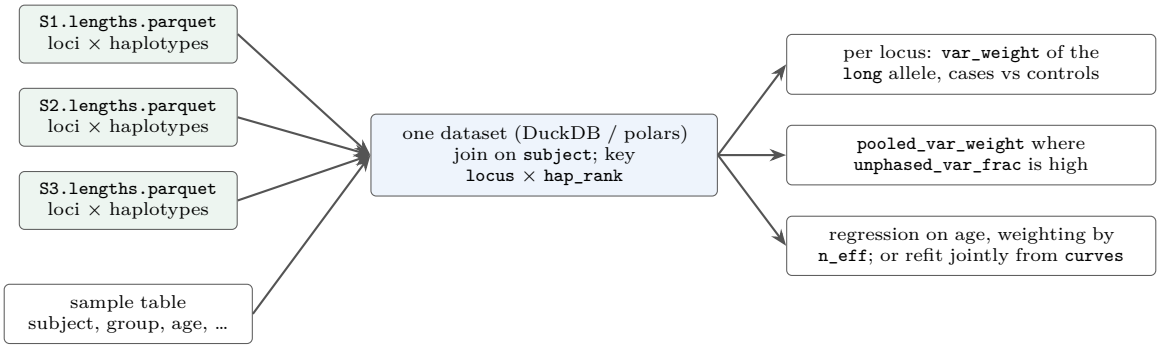


Figure 10: Group comparison. HiPhase's haplotype 1 means nothing across subjects, so rows are matched by `hap_rank`, or the pooled columns are used.

6 Statistical choices

choice	alternative	reason
learn $p(Y \mid L, x)$	learn $p(L \mid Y)$	a posterior bakes in the training prior over lengths; a likelihood transfers
softmax over Δ bins	parametric noise head	any error shape, learned from many stable reads
mixture loss over alleles	hard-assign reads	heterozygous loci and unphased reads used without guessing
measure Y between flank anchors	CIGAR inside repeat	indel placement inside repeats is arbitrary
soft haplotypes, w from hets only	hard phasing; fit w freely	free w is circular: reads would go where they look least unstable
ρ for unphased reads	50/50	the flank rule favours the shorter allele; fitting ρ absorbs it
cap w at $1 - \epsilon$	$w = 1$	one misphased read would become an outlier on the wrong haplotype
point mass + g	one distribution	interpretable stable fraction π; M0 nested at $\pi = 1$
skew-normal g (default)	nonparametric g	stable at modest depth; few numbers; nonparametric as a shape check
exclusion zone Δ_g	none	otherwise g collapses on L_0 and absorbs ± 1 bp noise
outlier starts, uniform EM floor	local starts only	a single expanded read must be reachable
selection $s(L)$	ignore	long molecules are under-sampled by the flank rule
no p-values; $d_{\ell\ell}$ and baselines	bootstrap LRT	the target is estimation; stable samples give the floor

Table 4: Choices, what they replace, and why.

Checks on simulated loci (`tests/test_tandemo.py`).

- A haplotype with 30% of molecules expanded is recovered at 0.23–0.41 across seeds, for both shapes. The stable haplotype stays at 0–0.06.
- One or two expanded reads among 40 per haplotype (+30 or +240 bp, phased or not) are captured with a HiFi-like noise model: $1 - \pi = 1/41$ or $2/42$, `n_support` = 1 or 2, log-likelihood gain 8–20. Stable loci give exactly 0.
- With all reads unphased, the two haplotypes are separated by length and $\rho \approx 0.5$.

The stable floor. A maximum-likelihood weight at a boundary is biased upward. With a noise model whose tails are too heavy, stable loci show $1 - \pi \approx 0.03$ on average at 40 reads per haplotype. Whether one or two deviant reads count as instability is decided by Network 1's tails: a sharp, well-calibrated noise model makes them decisive. Artefact reads (paralogs, chimeras) also produce outliers. Run Stage 2 on held-out stable samples and read instability as an excess over that baseline at matched depth and locus class.

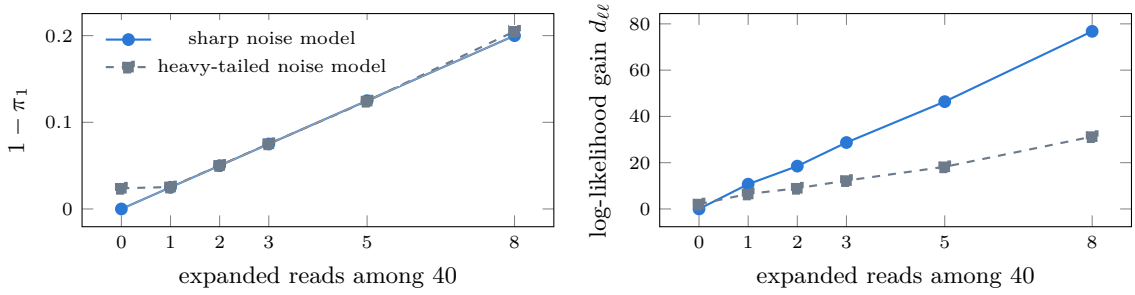


Figure 11: k reads expanded by 6–15 units among 40 phased reads of haplotype 1, median of 8 simulations. Left: the fitted variable weight tracks $k/40$ (grey line) under both noise models, so Stage 2 does not regularise single reads away. Right: the *evidence* for them depends on Network 1's tails. With a sharp model one read gains ≈ 11 log units and a stable locus gains 0. With heavy tails, one read gains about 6, and a stable locus already gains about 2.

7 Deferred

- **Truncated reads** (one good flank): a lower bound $L \geq c$, through a survival term in Network 1. They see expansions longer than reads can span.
- **Inside the repeat**: indel features, the motif-phase impurity spectrum, and unit decomposition by wrap-around alignment, as extra features or priors.
- **Allele-specific noise**: two curves per uncertain read, $\ell_i(L \mid x_1)$ and $\ell_i(L \mid x_2)$.
- **Uncertainty intervals** for π and g (profile likelihood or bootstrap over subjects), and a group-level model on the stored curves.

8 Usage

```
# 1. per sample, after minimap2 (somatic_sv/scripts/align_hifi.sh), HiPhase and TRGT
tandemo extract --bam s.bam --reference ref.fa --catalog trgt.bed.gz \
    --phased-vcf s.hiphase.vcf.gz --trgt-vcf s.trgt.phased.vcf.gz --out ex/s --threads 32

# 2. once, on stable sample(s)
tandemo train --input ex/stable1 ex/stable2 --out net1.pt          # PyTorch
tandemo train --input ex/stable1 --out noise.npz --empirical       # numpy baseline

# 3. per subject
tandemo fit --input ex/s --model net1.pt --out fit/s --threads 32 \
    [--shape nonparametric] [--write-curves]
```